

Brady Bangasser

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Systems engineer running production HPC clusters and researching high-performance and distributed computing, from scheduler internals to GPU-offload optimization at supercomputer scale.

EXPERIENCE

Research Assistant — Compiler Optimization, Iowa State University

Aug 2025 – Present

Ames, IA

- Built OMPRTF (OpenMP RunTime Fabric) for Iowa State's SWAPP lab, a custom OpenMP runtime and dynamic-analysis pipeline that applies aggressive, conditional optimizations to GPU-offload programs by lowering them to LLVM IR.
- Wrote an LLVM optimization pass that breaks opaque OpenMP runtime calls into discrete IR instructions, pruning redundant host-GPU data movement without destabilizing kernels.
- Built a dynamic profiling library that correlates OpenMP target logs with IR metadata to surface duplicate transfers and unused memory automatically; initial testing showed ~10% performance gains from the profiling-guided optimizations.
- Co-developed an LLVM-based tool that restructures disk I/O access patterns to speed up data loading for large ML/AI workloads.
- Analyzed Darshan I/O logs from Iowa State's Nova cluster and the Polaris supercomputer to locate and resolve system-level data-loading bottlenecks.

System Administrator, Iowa State University

Apr 2025 – Present

Ames, IA

- Operate and maintain a production HPC cluster with SLURM, Kubernetes, Active Directory, Ansible, Prometheus, and Apptainer, keeping it secure and available for a large user base.
- Built diagnostic and monitoring workflows that isolate hardware degradation, network bottlenecks, and software-stack faults before they take nodes offline.
- Built a TUI stress-testing tool in C that dynamically schedules HPL (LINPACK) jobs and watches live metrics to keep runs inside safe thermal and power limits, protecting node health while pushing for peak results.
- Automated node provisioning and recovery with Ansible so failed or new nodes rejoin service quickly.
- Integrated NCCL tests into the tool to validate multi-GPU InfiniBand bandwidth and surface interconnect problems early.
- Implemented ResMemPerGPU in a customized SLURM scheduler (C) to stop CPU jobs from consuming all memory on GPU nodes, protecting GPU workloads, with Makefile-driven builds and validation.
- Improved researcher accessibility by supporting user applications through OpenOnDemand and streamlining common workflows.

Research Assistant — Machine Learning, Iowa State University

Oct 2024 – May 2025

Ames, IA

- Designed and trained object-classification models with PyTorch and OpenCV, tuning data preprocessing and tensor operations to cut training latency without losing accuracy.
- Tested perception models on self-driving vehicles to catch false and failed classifications.

Research Assistant — High Performance Computing, Bethel University

Aug 2023 – Aug 2024

Saint Paul, MN

- Built and configured two bare-metal HPC clusters from scratch: networking, custom DNS and routing, user management, and SLURM job scheduling with VMware and Debian.
- Implemented a wildfire simulation in OpenMP, CUDA, C, and Fortran to benchmark cluster performance across heterogeneous compute.
- Wrote a lightweight daemon and control program to manage per-node status lighting across the clusters.

EDUCATION

M.S. Computer Science, Iowa State University

Aug 2025 – May 2027

GPA: 4.00

- Concurrent B.S./M.S. program
- Coursework: Advanced Topics in High Performance Computing, Distributed Systems and Middleware, Data Security in Machine Learning, Network Design and Security

B.S. Computer Science, Iowa State University

Aug 2024 – Dec 2026

GPA: 3.79 · Dean's List · Minors in Cybersecurity and Data Science

- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Rust, Bash, CUDA, Fortran

Infrastructure & Platform: Kubernetes, SLURM, Apptainer, Linux/Unix, Git

Compilers & HPC: LLVM, OpenMP, OpenMPI, PyTorch

INTERESTS

Aviation (flying small aircraft), rock climbing